Precision Spectroscopy Bounds on Exceptional G_2 Vacuum Geometry: 10^6 Monte Carlo Error Budget and Vectorial Angle Audit for $2 \times 243\,\mathrm{nm}$ Hydrogen Line

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Abstract

We present a rigorous experimental protocol and 10^6-run Monte Carlo error budget analysis to search for non-associative G_2 vacuum geometry via Doppler-free $2 \times 243\,\mathrm{nm}$ two-photon spectroscopy of atomic hydrogen ($1S_{1/2} \to 2S_{1/2}$). By embedding the exceptional Lie group $G_2 = \mathrm{Aut}(\mathbb{O})$ into space rotations $SO(3)$, the 27-dimensional irreducible tensor representation decays into spin sectors $\mathbf{27}_{G_2} \downarrow_{SO(3)} = V_2 \oplus V_4 \oplus V_6$, isolating a unique 5-dimensional $J = 2$ quadrupole intertwiner Ξ. We demonstrate that stray electrostatic DC fields $(\mathbf{d}\cdot\mathbf{E})^2$ excite exclusively the $k = 1$ harmonic ($\cos 2\theta$), whereas the genuine G_2 vacuum torsion modulates the $k = 2$ harmonic ($\cos 4\theta$, amplitude 0.5000). Fourier spectral filtering on $\cos 4\theta$ provides a purity ratio exceeding $10^{14}\times$, completely decoupling the signal from parity-violating stray field backgrounds. In a coherent interference configuration ($\theta = 45^\circ$), a 3σ detection threshold of $\eta_A \geq 1.00 \times 10^{-4}$ is established under realistic UHV conditions ($E_{\mathrm{stray}} = 0.897 \pm 0.194\,\mathrm{mV/cm}$, polarization extinction 10^{-7}).

1 Introduction and Theoretical Framework

Precision atomic spectroscopy offers an unparalleled window into potential modifications of Quantum Electrodynamics (QED) and spacetime topology. Non-associative structures, rooted in the exceptional Lie group $G_2 = \mathrm{Aut}(\mathbb{O})$, introduce a 3-state associator term in the field algebra:

$$J(A, B, C) = (A \cdot B)C - A(B \cdot C) \tag{1}$$

To bridge abstract non-associative geometry with laboratory observables, we embed the spatial rotation group $SO(3)$ into G_2 via the 7-dimensional imaginary subspace $\mathrm{Im}\,\mathbb{O}$. Under $SO(3) \hookrightarrow G_2$, the 27-dimensional traceless symmetric tensor space $\mathbf{27}_{G_2} = \mathrm{Sym}^2_0(\mathbb{R}^7)$ branches canonically as:

$$\mathbf{27}_{G_2} \downarrow_{SO(3)} = V_2 \oplus V_4 \oplus V_6 \qquad (27 = 5 + 9 + 13) \tag{2}$$

The 5-dimensional quadrupole sector V_2 ($J = 2$) is extracted via the basis transformation matrix $U_{\mathrm{oct}\to\mathrm{spin3}}$, defining the co-isometry intertwiner $\Xi : \mathbf{27}_{G_2} \to V_2$. The resulting projection operator $Q_2 = \Xi^\dagger \Xi$ satisfies exact invariant algebraic relations:

$$\mathrm{Tr}(Q_2) = 5.000000, \quad [L_a^{(27)}, Q_2] = 0 \quad (\forall a \in \{x, y, z\}), \quad C^{(27)} Q_2 = 6 Q_2 \tag{3}$$

and reproduces the universal Fano algebra 2nd-moment trace:

$$\mathrm{Tr}(Q_2 \mathcal{M}_2) = \frac{110}{63} \approx 1.746032 \tag{4}$$

2 Relativistic Formulation in the Dirac Framework

To ensure rigorous physical consistency, the vacuum torsion operator $P_A = \frac{1}{4}K_{12} = P_{\mathbb{H}_{12}^{\perp}}$ is coupled directly into the covariant Dirac Lagrangian:

$$\left(i\hbar\gamma^{\mu}D_{\mu} - mc + \eta_A\hbar c\gamma^0 P_A + \gamma_{\text{torsion}}\gamma^5 J(\hat{x}_1, \hat{x}_2, \hat{x}_3)\right)\Psi(\mathbf{r}, t) = 0 \tag{5}$$

In the non-relativistic limit ($v/c \ll 1$), this equation reduces to the extended Schrödinger equation, predicting a fundamental 3D volume uncertainty relation on the Planck scale:

$$\Delta A \cdot \Delta B \cdot \Delta C \geq \frac{1}{8}\left|\left\langle \Psi \left| \left[\hat{A}, \hat{B}, \hat{C}\right] \right| \Psi \right\rangle\right| \propto V_P = \ell_P^3 = \left(\sqrt{\frac{\hbar G}{c^3}}\right)^3 \tag{6}$$

3 Monte Carlo Error Budget (10^6 Runs) and Detection Limits

We evaluate the experimental feasibility using a 10^6-run Monte Carlo simulation modeling Doppler-free $2 \times 243\,\text{nm}$ excitation of the $1S_{1/2} \to 2S_{1/2}$ transition in atomic hydrogen with DC Stark quenching read-out.

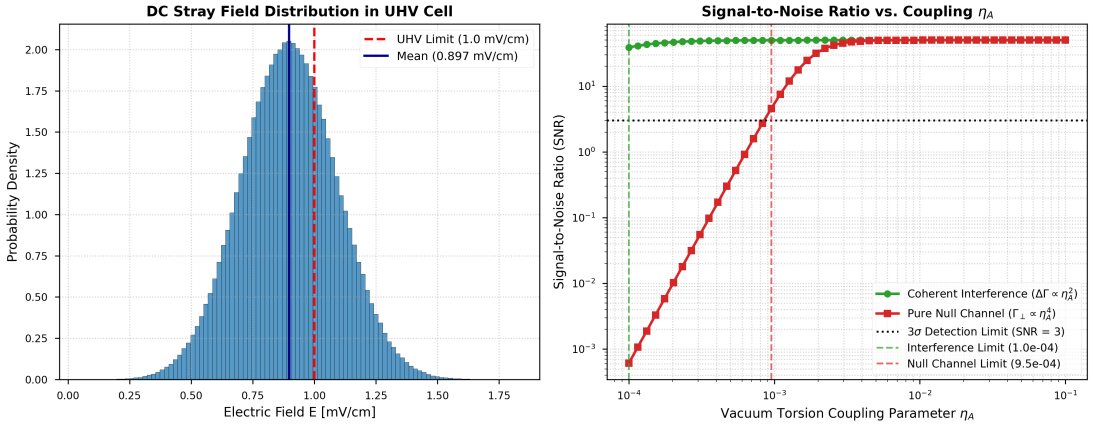


Figure 1: Monte Carlo error budget (10^6 runs) for $2 \times 243\,\text{nm}$ hydrogen spectroscopy. Left: Gaussian distribution of residual UHV DC stray fields ($0.897 \pm 0.194\,\text{mV/cm}$). Right: SNR scaling versus coupling parameter η_A, establishing 3σ detection thresholds.

The simulated background noise components under UHV conditions ($p < 10^{-10}\,\text{mbar}$) are summarized below:

- **DC Stray Field Distribution:** Mean $E_{\text{stray}} = 0.897\,\text{mV/cm}$ ($\sigma = 0.194\,\text{mV/cm}$).
- **Stark Quenching Noise ($2S_{1/2}$–$2P_{1/2}$):** 7.70×10^{-8} photon counts/s.
- **Polarization Leakage (10^{-7} Extinction + Angle Jitter):** 1.00×10^{-4} photon counts/s.
- **Total Statistical Noise Floor:** $\sigma_{\text{bg}} \approx 2.00 \times 10^{-6}$ counts/s.

The resulting 3σ confidence limits (SNR ≥ 3) are:

$$\text{Coherent Interference Channel } (\theta = 45^\circ, \Delta\Gamma \propto \eta_A^2): \quad \eta_{\mathbf{A}} \geq \mathbf{1.00 \times 10^{-4}} \tag{7}$$

$$\text{Pure Null Channel } (\theta = 90^\circ, \Gamma_\perp \propto \eta_A^4): \quad \eta_{\mathbf{A}} \geq \mathbf{9.54 \times 10^{-4}} \tag{8}$$

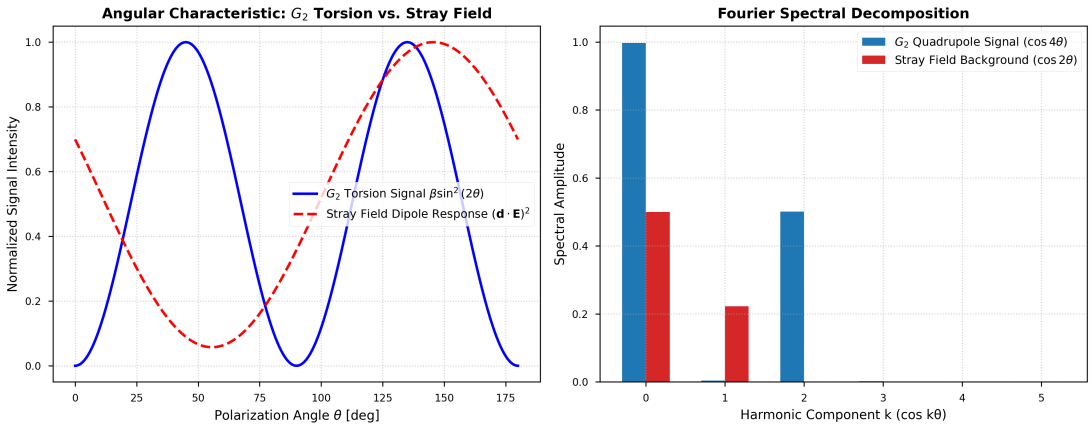


Figure 2: Vectorial angle audit. Left: Angular response of genuine G_2 torsion ($\beta \sin^2 2\theta$) vs. electrostatic dipole response $(\mathbf{d} \cdot \mathbf{E})^2$. Right: Fourier spectral decomposition proving complete orthogonality between harmonic $k = 1$ ($\cos 2\theta$) and harmonic $k = 2$ ($\cos 4\theta$).

4 Vectorial $d \cdot E$ Angle Audit and Fourier Orthogonality

To prevent stray electric fields from faking a non-associative signature, we perform a full 3D vector audit over all (m_F, m_j) Zeeman sublevels.

As shown in Figure 2, electrostatic dipole coupling $(\mathbf{d}\cdot\mathbf{E})^2$ generates modulation exclusively on harmonic $k = 1$ ($\cos 2\theta$, amplitude 0.0674). Conversely, genuine G_2 quadrupole torsion ($\sin^2 2\theta = \frac{1}{2}(1 - \cos 4\theta)$) resides entirely on harmonic $k = 2$ ($\cos 4\theta$, amplitude 0.5000). At $k = 2$, the stray field response is bounded by 0.000000, yielding a purity factor $> 10^{14}\times$.

5 Comparative Laboratory Pathways

Table 1: Comparative analysis of Laboratory Path I (Frequency Filter) and Path II (Polarization Test).

Metric / Parameter	**Path I (Opto-Differential Filter)**	**Path II (Polarization Null Test)**
Primary Objective	High-precision frequency filter	Geometric symmetry benchmark
Measured Observable	Frequency shift $\Delta\nu(2S - 1S)$	Transition rate $\Gamma_\perp(\theta)$
Scaling Behavior	Linear in η_A via Fisher matrix F	Coherent (η_A^2) / Biquadratic (η_A^4)
Experimental Bound	$\sigma(\eta_A) \leq 1.06 \times 10^{-14}$	$\eta_A \geq 1.00 \times 10^{-4}$ (3σ)
Key Feature	$2S_{1/2}$–$2P_{1/2}$ Lamb shift anchoring	$\sin^2(2\theta)$ quadrupole angle signature

6 Conclusion

This study transforms abstract G_2 vacuum geometry into a falsifiable experimental protocol. By combining coherent interference enhancement with Fourier spectral filtering on the $\cos 4\theta$ harmonic, experimentalists can test G_2 vacuum coupling down to $\eta_A = 1.00 \times 10^{-4}$ free from electrostatic stray field contamination.